Parametric & Sequences Test

Cram sheet

Wed, no calculator: 5 parametric curve questions + 1 sequence question. **Fri, calculator:** 7 sequences & series + 1 projectile. 43 points.

Memorize: not given on the test

(1) $\sum ca_k = c\sum a_k$

(2), (3) $\sum(a_k \pm b_k) = \sum a_k \pm \sum b_k$

(4) $\sum_{k=j+1}^{n} a_k = \sum_{k=1}^{n} a_k - \sum_{k=1}^{j} a_k$ (j = start -1)

(5) $\sum_{k=1}^{n} 1 = n$ (6) $\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$

(7) $\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$

(8) $\sum_{k=1}^{n} k^3 = \left[\frac{n(n+1)}{2}\right]^2$

$0! = 1,\ 1! = 1$ • $\sin^2 t + \cos^2 t = 1,\ 1 + \tan^2 t = \sec^2 t,\ 1 + \cot^2 t = \csc^2 t$ • Given: $x = (v\cos\theta)t,\ y = (v\sin\theta)t - 16t^2 + h$

Find a sum (Friday)

1. Lower index not 1: **fix it with (4) first**. Start at $k = 6$ means $j = 5$.
2. Split with (2), (3); constants out with (1).
3. Write each formula from memory with n plugged in. **This line is required before the calculator.** The Σ button is blocked in test mode.

Ex: $\sum_{k=6}^{20} 4k^2 = 4\left[\frac{20(21)(41)}{6} - \frac{5(6)(11)}{6}\right] = 11260$ $\sum_{k=1}^{10}(k^3+1) = \left[\frac{10(11)}{2}\right]^2 + 10 = 3035$

Binomial Theorem: Pascal's triangle (Friday)

1
1 1
1 2 1
1 3 3 1
1 4 6 4 1
1 5 10 10 5 1
1 6 15 20 15 6 1
1 7 21 35 35 21 7 1
1 8 28 56 70 56 28 8 1

- Outsides 1; inside = sum of the two above. Row n for power n (up to about 7 or 8).
- Blanks first: first term's powers **down** $n \to 0$, second term's **up** $0 \to n$. Fill with row n, simplify.
- The sign goes with its term; signs alternate only when the second term is negative. Use Pascal's triangle: multiplying it out loses credit.
- $(2x)^4 = 16x^4$, not $2x^4$.

Ex: $(2x-3)^4 = 16x^4 - 96x^3 + 216x^2 - 216x + 81$

Sequences

- **Terms:** plug in $n = 1, 2, 3, \ldots$ Answer is a list with commas, reduced, no ellipsis.
- ***n*th term:** write 1 2 3 4 above; top and bottom separately; first term positive $(-1)^{n+1}$, negative $(-1)^n$; odds $2n-1$, squares n^2, powers b^{n-1}. Check $n = 1$. Ex: $\frac{1}{3}, -\frac{2}{9}, \frac{3}{27}, \ldots \to a_n = (-1)^{n+1}\frac{n}{3^n}$
- **Recursive:** first term given; the formula starts where it says ($n \geq 2$); a_{n-1} is the previous term.
- **Write out a sum:** first few terms, ellipsis, last term; don't add. **Express as Σ:** lower index, upper index, general term, same letter. Check $k =$ lower index gives the first term.

How far to simplify (every rectangular answer)

- **Whole graph shows** (circle, ellipse, both branches): done once t is gone, e.g. $\frac{x^2}{4} - y^2 = 1$. Solving for y is also right, only with $\pm$.
- **Part of it, a domain can cut it** (piece of a parabola, one left-right branch): add "for x in $[a,b]$", e.g. $\frac{x^2}{4} - y^2 = 1$ for x in $[2,\infty)$.
- **Part of it, a domain can't** (top or bottom of an up-down hyperbola): solve for y, keep one sign, e.g. $y = 2\sqrt{\frac{x^2}{9} + 1}$.
- **Small piece of a branch:** solve for y *and* add "for x in". Never "for y in" or "$y \geq$".
- Question says **SIMPLIFIED**: all the way, y alone ($y = -\frac{1}{3}x^2$). No "simplify": any form with t gone; don't multiply out.

Parametric curves (Wednesday)

1. **t-chart first.** Brackets: plug in both ends, solid dots. Parentheses: values inside.
2. Solve for t where it gives **one answer** (no $\pm$); substitute into the other.
3. Domain: smallest and largest x in the chart, "for x in $[a,b]$". Watch turning points ($x = t^2 \geq 0$, $x = \sqrt{t} \geq 0$).
4. Any form is fine once t is gone; no multiplying out. **SIMPLIFIED** means y alone. No domain written means all real x.
5. Direction arrows **only when the question asks**: lowest t to highest. End arrowheads only when the curve keeps going. Asymptotes optional, dashed if drawn.

What the t-chart shows	**Final form** (whole graph: done; domain can cut it: "for x in"; can't: solve for y, one sign)
full circle / ellipse	$\frac{x^2}{9} + \frac{y^2}{25} = 1$, no domain
one hyperbola branch (restricted t)	solve for y, keep that branch's sign ($+\sqrt{}$ top, $-\sqrt{}$ bottom): $y = 2\sqrt{\frac{x^2}{9} + 1}$; never "$y \geq$"; any equivalent form
both branches (no restriction)	$\frac{x^2}{4} - \frac{y^2}{9} = 1$, done (or $y = \pm 3\sqrt{\frac{x^2}{4} - 1}$, never $+\sqrt{}$ alone)
one left-right branch	$\frac{x^2}{4} - y^2 = 1$ for x in $[2,\infty)$; never "for y in"
reciprocal $x = t + 2,\ y = \frac{1}{t+2}$	$y = \frac{1}{x}$; "for all $x \neq 0$" is good practice (it comes with the equation)

Two parametric representations

Give both $x =$ and $y =$ ($t = 4 - x$ is not parametric form). $y = (x-2)^2 + 1$: $\quad x = t,\ y = (t-2)^2 + 1 \quad$ and $\quad x = t + 2,\ y = t^2 + 1$, for t in $(-\infty, \infty)$ (or the restricted interval). Not in vertex form? Pick any shift $x = t + h$, substitute, simplify.

Projectile (Friday)

1. Equations with the numbers and the **degree symbol** (leaving it off is not okay); from the ground $h = 0$.
2. Degree mode, Graph, Parametric, $x_1(t)$, $y_1(t)$ exactly; write the setup. Path doesn't come down? Raise $t$max.
3. Max height: **trace** to the top, read y (Analyze, Maximum doesn't work on parametric). Time: $y \approx 0$, read t. Distance: x at that point. Trace step 0.01, then 0.001.
4. Thousandths, units in feet; "$y \approx$", "$t \approx$", "$x \approx$" are full answers: $y \approx 26.258$ ft, $t \approx 2.512$ s, $x \approx 126.707$ ft.

Where points get lost

- Writing only $+\sqrt{}$ for a whole hyperbola (leave it in "$= 1$" form or keep $\pm$). Leaving off the degree symbol.
- Arithmetic slips while simplifying: check one t-chart point in the final equation.
- Using a sum formula before fixing a lower index that isn't 1; skipping the substituted-formula line.
- $(2x)^n$ written as $2x^n$; the minus sign left off the second term.
- Wrong sign exponent on alternating sequences: check a_1.